PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. under the Panjab University Act VII of 1947 – enacted by the Govt. of India)

SYLLABI

Course Outline for
B.Voc. (Automobiles)

as per NEP 2020

w.e.f session 2024-25

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as per NEP 2020 w.e.f. session 2024-25

Semester-I

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
AM 101	Core Discipline	Elements of Automobile Technology-I	2	2	4	40	10	40	10	100
AM 102	Core Discipline	Automotive Electronics and Electric System	2	2	4	40	10	40	10	100
AM 103	Core Discipline	Automotive Tools	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English			2					50
B.Voc. MDSC 101	Multidisciplinary Course	Fundamentals of Information Technology[1]	3	-	3	60	15	-	-	75
SEC I	Skill Enhancement Course	SEC Options[2]			3					75
VAC I	Value-Added Course	VAC Options[3]			2					50
PW I	Project Work	Project Work-I & Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-II

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
AM 201	Core Discipline	Elements of Automobile Technology-II	2	2	4	40	10	40	10	100
AM 202	Core Discipline	Advanced Electronic and Auto Electric System	2	2	4	40	10	40	10	100
AM 203	Core Discipline	Automobile Physics	4	-	4	80	20	-	-	100
AEC / MIL	Ability Enhancement Course	English			2					50
B.Voc. MDSC 201	Multidisciplinary Course	Soft Skills and Personality Development[1]	3	-	3	60	15	-	-	75
SEC II	Skill Enhancement Course	SEC Options[2]			3					75
SIT 201*	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

Job Role - Assistant Auto Technician

* Summer Industrial Training of 2-4 weeks in a relevant Industry after 2nd Semester Examinations during summer break. Training report by the student to be submitted within one week of start of 3rd Semester. Viva-Voce examination to be held within 3 weeks of the start of 3rd semester.

** Minimum Credits required to get Certificate Course in Automobile = 48 + 4 #.

#Exit 1: Certificate Course - Student can exit after completion of 1st year having secured 48 Credits and will be awarded a UG Certificate provided they complete a vocational course/internship of 4 credits during the summer vacation of the first year, in addition to the 6 credits from skill enhancement courses earned during first and second semester.

1. Multidisciplinary Courses (MDSC): Annexure I
2. Skill Enhancement Courses (SEC): Annexure II
3. Value Added Courses (VAC): Annexure III

ESSENTIAL COMPONENTS OF COURSE OF B.VOC (Automobiles) (SEMESTERS I & II)

PROGRAMME OUTCOMES

PO1. Understanding of the basic concepts and principles in automobiles.

PO2. Knowledge about automobile component both mechanical and electrical/electronic.

PO3. Understanding and practical experience of the automotive tools and equipment's.

PO4. Understanding of concepts and principles in automotive electronic system.

PO5. Adequate Professional Skills and Competencies in measuring instruments.

PO6. Knowledge about operation/working of engines along with its components.

Semester I

	COURSE TITLE	ELEMENTS OF AUTOMOBILE TECHNOLOGY-I
1	**Course Code**	AM - 101
2	**Typology of Course**	Core Discipline
3	**Course Duration/ LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of practical for each unit
4	**Credits**	4 Credits [2+2]
5	**Max. Marks and Time**	100 Marks (80 TH/PR + 20 IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6	**Job Role**	Assistant Auto Technician
7	**Course Objectives**	This course is designed to understand the concept of automobile. Knowledge about engine working and role of various components of engine in performance of automobiles. Also this course enhance deep and vigorous understanding of fuel supply system, cooling system, lubrication system, transmission system and steering geometry of a vehicle and other major components of automobile.
8	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Understand the concept of automobile and its major components. CO2. Knowledge about working of engine and role of various components of engine in performance of automobiles. CO3. Enhance understanding of fuel supply system, cooling system, lubrication system, transmission system and steering geometry of a vehicle. CO4. Recognize components of automobile CO5. Knowledge about basics of automobile air conditioner and emission system.
9	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11 **Course Content**

UNIT– I

Introduction to automobile, Classification of automobiles, Major components of automobile – Basic structure, power plant, transmission system, auxiliaries and controls.

UNIT– II

Introduction to Engines, Engine terminology, Basic engine parts and their uses

UNIT– III

Introduction to fuel system of petrol and diesel engines, Cooling system, Lubrication, Clutch, Gearbox, Brakes and Suspension system

UNIT-IV

Introduction to chassis, Wheels and Tyres, Propeller shaft, Rear axle assembly, Steering and Front axle, Automobile Emission Control, Heating and Air Conditioning System

Practical

1. Study and identification of automobile components, engine parts and their uses.
2. Introduction and demonstration of clutch and gearbox.
3. Study working of brakes (Hydraulic and Mechanical) and suspension of automobiles.
4. Familiarization with fuel system of both petrol and diesel engine, simple carburetor, ignition system of petrol engine.
5. Understanding working of cooling system (Air cooling and Water cooling) and lubrication system in automobiles.
6. Study of automotive heating and air conditioning system

12 **Suggested Resources for Reading**

1. Basic Automobile Engineering by C P Nakra Published by Dhanpat Rai and Sons, Jalandhar.
2. Automotive Mechanics by Wiliam H. Crouse, Donald L. Angkin Published by Tata McGraw-Hill publishing Pvt. Limited, New Delhi

Note: Latest edition of text books may be used.

	COURSE TITLE	AUTOMOTIVE ELECTRONICS AND ELECTRIC SYSTEM
1	**Course Code**	AM - 102
2	**Typology of Course**	Core Discipline
3	**Course Duration/ LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4	**Credits**	4 Credits [2+2]
5	**Max. Marks and Time**	100 Marks (80 TH/PR + 20 IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6	**Job Role**	Assistant Auto Technician
7	**Course Objectives**	This course is designed for comprehensive study of AC and DC circuits that are required in automobile electrical and electronic systems. Students will also learn about construction, working of transformer and principle of different indicating instruments. Knowledge about generation of alternating electromotive force, concept of EMF, root mean square and average value of AC current also included in this course.
8	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Understand concept of AC and DC circuits that are required in automobile electrical and electronic systems. CO2. Theoretical and practical exposure about construction and working of transformer. CO3. Knowledge about different indicating instruments. CO4. Understand concept of EMF in detail. CO5. Practical exposure of multimeter to measure AC and DC voltage and currents in the circuit.
9	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11 **Course Content**

UNIT– I

Electricity, Elements of electricity, Ohm's Law, Electric circuits, DC measuring instruments, Action of electric current, Voltage generation, Electrochemical cells

UNIT– II

Introduction to magnetic and electric fields, Faraday's law of electromagnetic induction, Self and Mutual inductance, Coupling coefficient, Voltage regulation and its applications

UNIT– III

Basic principle and classification of indicating instruments, Digital and analogue multimeter, Measurement of power energy, Resistance control and protection device, Circuit breaker, Fuse, MCB

UNIT-IV

Concept of EMF, Induced EMF and its types, Concept of resonance in series LCR circuit, Principle, construction and working of transformer, Core type and shell type transformers

Practical

1. Use of multimeter to measure AC and DC voltage and currents in the circuit.
2. Study of voltage, current and resonance in different circuits using Ammeter, Voltmeter, Galvanometer.
3. Study of circuit breaker and MCB used in automobiles, transformer's working, construction and its applications
4. Verification of ohm's experimentally.
5. Study the function of CRO.

12 **Suggested Resources for Reading**

1. Fundamentals of electrical and Electronics Engineering by B.L. Theraja, S. Chand and company, New Delhi.
2. Basic Electronic and electrical engineering by Bhattacharya S.K., Pearson Education
3. Principles of electrical engineering by Gupta B.R., S. Chand and Company, New Delhi.

Note: Latest edition of text books may be used.

	COURSE TITLE	AUTOMOTIVE TOOLS
1	**Course Code**	AM - 103
2	**Typology of Course**	Core Discipline
3	**Course Duration/ LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4	**Credits**	4 Credits [2+2]
5	**Max. Marks and Time**	100 Marks (80 TH/PR + 20 IA) i) The exam of this course will be of 3 hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6	**Job Role**	Assistant Auto Technician
7	**Course Objectives**	This course is designed to provide students with deep understanding of automotive tools. They will also learn about the working of various power operated tool. Knowledge about threads, fasteners, nut, bolts, bearings and belts. Concept of service information and measuring instruments are also included in this course.
8	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Understand automotive tools and power operated instruments. CO2. Deep knowledge about threads, fasteners, nut, bolts, bearings and belts. CO3. Practical exposure to operation of power operated tools. CO4. Understand service information and measuring instruments. CO5. Practical experience to use vernier caliper and screw gauge.
9	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11 **Course Content**

UNIT– I

Introduction to automotive tools, Hand tools, Wrenches, Screw driver, Pliers, Hammers and Mallets, Punches, Metal working tools, Hacksaws, Drills, Taps and Dies, Power tools – Electric, Hydraulic and Compressed air, Bench vise

UNIT– II

Fasteners, Threaded Fasteners, Thread size and designations, Washers, Torqueing Bolt, Removing of broken bolts, Gaskets, Adhesives, Sealers, Seals, Hoses and Lines, Belts and Bearings

UNIT– III

Introduction to service information, Steps in engine services, Safety tips for using hand tools, Safety tips for technicians, Engine service tools and equipment

UNIT-IV

Measuring instruments, Meters, testers and analysers, Testing safety rules, Using welding equipment, Shop safety, Fire prevention, Safety in hoisting a vehicle, Disposing of automobile wastes

Practical

1. Study and identification automotive tools
2. Introduction and demonstration of Fasteners, Washers, Torqueing Bolt, Gaskets, Adhesives, Sealers, Seals, Hoses and Lines, Belts and Bearings
3. Study operation of power operated tools
4. Calibration of vernier caliper, height/depth gauge and screw gauge.
5. To measure length, breath, thickness, depth, height, with vernier calipers

12 **Suggested Resources for Reading**

1. Automobile Engineering III by P.S. Gill Published by S.K. Kataria and Sons, New Delhi
2. Basic Automobile Engineering by C P Nakra Published by Dhanpat Rai and Sons, Jalandhar.
3. Automotive Mechanics by Wiliam H. Crouse, Donald L. Angkin Published by Tata McGraw-Hill Publishing Pvt. Limited, New Delhi

Note: Latest edition of text books may be used.

Semester II

	COURSE TITLE	ELEMENTS OF AUTOMOBILE TECHNOLOGY-II
1	**Course Code**	AM - 201
2	**Typology of Course**	Core Discipline
3	**Course Duration/ LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4	**Credits**	4 Credits [2+2]
5	**Max. Marks and Time**	100 Marks (80 TH/PR + 20 IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6	**Job Role**	Assistant Auto Technician
7	**Course Objectives**	The students will understand how engines are classified either on the basis of valve arrangement, number of strokes, cylinder arrangement, fuel used, engine cycles etc. Students should be able to understand about the working of petrol and diesel engines along with their merits and demerits. They will also gain knowledge of various components of engine and also how they affect the performance of an engine**.**
8	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1.Knowledge about engine classification in detail. CO2.Understand working of petrol and diesel engine along with its merits and demerits. CO3.Practical exposure of major components used in engines. CO4. Knowledge about cylinder block, cylinder head, piston, piston rings in detail. CO5.Deep understanding of engine performance affected by size and material of engine components.
9	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11 **Course Content**

UNIT– I

Classification and types of engines based on engine cycles, Number of strokes. Fuel use, Number and Arrangement of cylinder, Valve arrangement, Type of valve. Special type engine- Square engine, Wankel engine, Atkinson cycle engine, Miller cycle engine and Electrical engine

UNIT– II

Petrol engine: - Introduction, merits and demerits. Diesel engine: - Introduction, merits and demerits. Valve timing of two stroke and four stroke engine

UNIT– III

Engine construction I: Cylinder blocks, Cylinder head, Gasket, Piston, Piston ring, Cylinder liner, Manifold, Piston pumps, Connecting rods

UNIT-IV

Engine construction II: Vibrant on damper, Camshaft, Engine bearings, Timing belt/chain, Balanced weight, Flywheel, Crankcase, Crankshaft and Valve

Practical

1. Introduction and demonstration of engine cycles, number of strokes, number of cylinder and its arrangements.
2. Familiarization of petrol and diesel engines used in automobiles
3. Study and demonstration of engine construction and components, i.e., piston, piston ring, cylinder block, cylinder head, cam shaft, crank shaft, vibrant damper manifold and timing chain/belt.
4. Study of construction and working principles of Three phase induction motor, Synchronous machines, DC machines

12 **Suggested Resources for Reading**

1. Basic Automobile Engineering by C P Nakra Published by Dhanpat Rai and Sons, Jalandhar.
2. Automotive Mechanics by Wiliam H. Crouse, Donald L. Angkin Published by Tata McGraw-Hill publishing Pvt. Limited, New Delhi

Note: Latest edition of text books may be used.

	COURSE TITLE	ADVANCED ELECTRONIC AND AUTO ELECTRIC SYSTEM
1	**Course Code**	AM - 202
2	**Typology of Course**	Core Discipline
3	**Course Duration/ LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4	**Credits**	4 Credits [2+2]
5	**Max. Marks and Time**	100 Marks (80 TH/PR + 20 IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6	**Job Role**	Assistant Auto Technician
7	**Course Objectives**	The objective of this course is to make students understand about construction and working principle of DC machines, three phase induction motor and synchronous motor. This will also enable the students to understand about diodes, number system, microprocessors, flip flops, amplifiers, oscillators and multi vibrators. Knowledge about major electrical components used in automobiles.
8	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Knowledge about working of DC machines, three phase induction motor and synchronous motor. CO2.Practical exposure to study power and energy measurement using energy meter. CO3.Understand working of microprocessors, operational amplifiers, oscillators, multivibrators. CO4. Exposure to study characteristics of BJT and working of LCR circuit practically. CO5. Understand working of engine starter, generator/alternator, ignition coil and other electrical components.
9	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11 **Course Content**

UNIT– I

DC machines (motors and generators), Three phase induction motor, Synchronous machines (motors and generators): construction and working principles

UNIT– II

Principle of operation, characteristics and applications of PN junction diode, Rectifiers, Zener diodes, bipolar junction transistors, Principle of operation, characteristics and applications of FET

UNIT– III

Binary, Octal and hexadecimal number system and its arithmetic operations, Introduction to microprocessors, operational amplifiers, oscillators, multivibrators, Testing ICs, Micro phone and loud speakers, LEDs, LCDs.

UNIT-IV

Introduction to engine starter, generator/alternator, ignition coil and other electrical components (fuses, relays, lighting system) used in automobiles

Practical

1. Study characteristics of PN junction diode.
2. Study of Full and Half wave Rectifiers.
3. Study of V-I characteristics of Zener diodes.
4. Characteristics of BJT and working of LCR circuit.
5. Study power and energy measurement using energy meter.

12 **Suggested Resources for Reading**

1. Fundamentals of electrical and Electronics Engineering by B.L. Theraja, S. Chand and company, New Delhi.
2. Basic Electronic and electrical engineering by Bhattacharya S.K., Pearson education
3. Principles of electrical engineering by Gupta B.R., S. Chand and company, New Delhi.

Note: Latest edition of text books may be used.

	COURSE TITLE	AUTOMOBILE PHYSICS
1	**Course Code**	AM - 203
2	**Typology of Course**	Core Discipline
3	**Course Duration/ LP (Lectures) (Practical)**	60 hours of lectures for Theory (1 credit = 15 hours of theory learning based on 4 hrs./week) (1 Semester = 15 weeks)
4	**Credits**	4 Credits
5	**Max. Marks and Time**	100 Marks (80 TH/PR + 20 IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6	**Job Role**	Assistant Auto Technician
7	**Course Objectives**	The course under automobile physics has been designed to give the basic knowledge about the displacement, velocity and acceleration of automobiles. They also learn about kinetic energy potential energy and fraction associated with the automobiles. Fundamentals of AC, DC, resistance, conductance and gates (AND OR, NOT) are also included in this course.
8	**Course Outcomes**	On successful completion of the course, the students will be able to CO1. Understand layout of electrical system in Automotive. CO2. Knowledge about the displacement, velocity and acceleration. CO3. Exposure to binary number system. CO4. Knowledge about colour code for carbon resistors CO5. Deep understanding of capacitors (Parallel and Series)
9	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11 **Course Content**

UNIT– I

Difference between Alternating current and Direct current, Layout of electrical system in Automotive, Automotive wiring and its forms, Kirchoff's Law

UNIT– II

Newton's Law of Motion, Distance, Displacement, Velocity, Speed, Acceleration, Variable force, Kinetic and potential energy, Work energy theorem, Static and kinetic friction, Rolling friction, Lubrication

UNIT– III

Charge and its properties, electric field, potential, capacitor, parallel plate capacitors, capacitance, combination of capacitor in series and parallel, Resistance, Electrical resistivity, Conductivity, Resistor in series and parallel combination, Colour code for Carbon resistors

UNIT-IV

Binary number system, Addition and subtraction for binary numbers, OR gate, AND gate, NOR gate, NAND gate and XOR gate and their truth tables

12 **Suggested Resources for Reading**

1. Modern ABC of Physics +1 and +2 by A. Gupta, Modern Pub. Jalandhar.
2. JBD Physics +1 and +2 by S.K. Gupta, K.L. Gosian and B.S. Satyal

Note: Latest edition of text books may be used.